

Shilpitha Chowdary Thokala

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EDUCATION

Johns Hopkins University

M.S.E. in Robotics (Expected May 2027)

Coursework: Deep Learning, Computer Vision, Learning Based Vision

Baltimore, MD

Aug 2025 – Present

Amrita Vishwa Vidyapeetham

B.Tech, Automation and Robotics Engineering — CGPA 9.02/10

Coimbatore, India

2021 – 2025

RESEARCH EXPERIENCE

International Institute of Information Technology (IIIT)

Research Intern — Autonomous UAVs

Hyderabad, India

Jan 2025 – Jul 2025

- Built a vision-based obstacle-detection system for GPS-denied indoor UAV flight: collected and annotated 2,500+ images, trained CNN detectors, and deployed to embedded hardware for autonomous navigation.
- Implemented visual odometry for indoor localization and an MPC trajectory controller (Python/C++, ROS2, PX4), validated in Gazebo and across 10+ hardware flight tests (Docker, Linux, CMake).

Indian Institute of Technology (IIT)

Research Intern — Autonomous Mobile Robots

Palakkad, India

Jun 2024 – Aug 2024

- Benchmarked classical (A*, RRT, APF) and RL-based path planning algorithms in ROS2 for AMR navigation, with RL approach achieving highest success rate in cluttered environments. Achieved 97.8% pathfinding success rate in environments with dynamic obstacles and narrow corridors.

RESEARCH PROJECTS

Diagnosing View-Consistency Failures in Affordance Prediction Models

Spring 2026

Graduate course research project

- Diagnosed failure modes across two affordance pipelines (RoboPoint/CLIP, GAT/DINOv2) and showed the dominant failure axis is functional-part visibility rather than viewpoint: grasp confidence fell from 0.99 (part visible) to 0.56 (fully occluded).
- Proposed the model's internal grasp-confidence score as a training-free signal a downstream VLA policy can read to trigger active perception, requiring no retraining of the affordance backbone.

Multimodal Retrieval-Augmented Generation for Surgical Documentation

2026 – Present

Graduate research, ongoing

- Adapting a multimodal RAG pipeline to automate surgical operative-note generation from procedural video, as part of a research team working toward publication.

Probing Visual Affordance in Vision-Language-Action Encoders

Spring 2026

Research contributor — graduate course research project

- Characterized how part/affordance structure evolves across VLA fine-tuning stages (SigLIP $\rightarrow$ $\pi_0 \rightarrow \pi_{0.5}$) using per-layer PCA and subspace-projection analysis, identifying a stage-specific representation collapse invisible to supervised probing.

SELECTED PROJECTS

Neural Deblurring for Robust Image Stitching

Aug 2025 – Dec 2025

- Built a panorama-stitching pipeline pairing a fine-tuned Restormer deblurring model with SIFT/SuperPoint feature matching; cut reprojection error 34% versus a SIFT-only baseline on 200+ motion-blurred pairs.

Underwater Vehicle-Manipulator System

Aug 2024 – May 2025

- Modeled full dynamic coupling of an underwater vehicle-manipulator system in Simulink and designed a sliding-mode controller for trajectory tracking under hydrodynamic disturbances.

PUBLICATION & LEADERSHIP

- Publication:** VQ-VAE-based trajectory generation for multi-drone coordination, ICCA 2025.
- Course Assistant, Computer Vision, Johns Hopkins University (Spring 2026).
- Led a ROS2 workshop for 70+ students with the campus robotics club; E-Yantra national finalist (IIT Bombay) for an LQR self-balancing robot.

SKILLS

Programming: Python, C++, C, MATLAB.

ML / Vision: PyTorch, OpenCV, VLMs & VLA models, diffusion models, DINOv2, CNNs (YOLO, Faster R-CNN), SIFT/SuperPoint, SLAM.

Tools & Robotics: Docker, Linux, CMake, Git; ROS, ROS2, PX4, Gazebo.